

DATA PAPER

Long-term seedling and small sapling census data from the Barro Colorado Island 50 ha Forest Dynamics Plot, Panama

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Abstract

Tropical forests are well known for their high woody plant diversity. Processes occurring at early life stages are thought to play a critical role in maintaining this high diversity and shaping the composition of tropical tree communities. To evaluate hypothesized mechanisms promoting tropical tree species coexistence and influencing composition, we initiated a census of woody seedlings and small saplings in the permanent 50 ha Forest Dynamics Plot (FDP) on Barro Colorado Island (BCI), Panama. Situated in old-growth, lowland tropical moist forest, the BCI FDP was originally established in 1980 to monitor trees and shrubs ≥ 1 cm diameter at 1.3 m above ground (dbh) at ca. 5-year intervals. However, critical data on the dynamics occurring at earlier life stages were initially lacking. Therefore, in 2001 we established a 1-m² seedling plot in the center of every 5 × 5 m section of the BCI FDP. All freestanding woody individuals ≥ 20 cm tall and < 1 cm dbh (hereafter referred to as seedlings) were tagged, mapped, measured, and identified to species in 19,313 1-m² seedling plots. Because seedling dynamics are rapid, we censused these seedling plots every 1–2 years. Here, we present data from the 14 censuses of these seedling plots conducted between the initial census in 2001 to the most recent census, in 2018. This data set includes nearly 1 M observations of ~185,000 individuals of >400 tree, shrub, and liana species. These data will permit spatially-explicit analyses of seedling distributions, recruitment, growth, and survival for hundreds of woody plant species. In addition, the data presented here can be linked to openly-available, long-term data on the dynamics of trees and shrubs ≥ 1 cm dbh in the BCI FDP, as well as existing data sets from the site on climate, canopy structure, phylogenetic relatedness, functional traits, soil nutrients, and topography. This data set can be freely used for non-commercial purposes; we request that users of these data cite this data paper in all publications resulting from the use of this data set.

KEYWORDS

advanced regeneration, Barro Colorado Island (BCI), Forest Dynamics Plot (FDP), ForestGEO, Panama, regeneration niche, seedling bank, seedling growth, seedling recruitment, seedling survival, species coexistence, tropical forest

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The complete data set is available as Supporting Information. The data set is also available in Dryad at <https://doi.org/10.5061/dryad.05qfttf85>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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